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MediPath AI is a Zero-Runtime Serving Divergence and Procedural Cost Transparent Clinical Triage and Healthcare Navigation System with multiple labels.

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***Abstract*— MediPath AI is an intelligent clinical decision support and hospital recommendation system to enhance healthcare navigation for healthcare providers in resource constrained environment like Tier-2 and Tier-3 cities. It will address some of the critical issues such as unreliable symptom checkers among hospitals, lack of integrated information on hospital services and unclear treatment cost borne by patients. It is an AI system for multi-label classification which was trained using MedP_AI_V2.0 model, a 56-label model on 104,291 curated clinical records with a leakage-free data partitioning strategy. It has a bidirectional symptom assessment, an adaptive counter-questioning, a differential condition disambiguation, a safe-deferral pathway to a GP for unsupported or uncertain conditions and handling of negative responses. Deterministic emergency and oncology overrides enable safer triage, but do not make a definitive diagnosis. The hospital recommendation engine lists those hospitals which have been recommended by specialists, and offers falls-back options, and its cost estimation module helps compare the cost of treatment, which is budget, mid-tier and premium, along with NABH accreditation adjustments. Evaluated on 15,764 held-out records, the model achieved Top-1, Top-3, and Top-5 hit rates of 88.19%, 99.03%, and 99.80%, respectively, with a Micro F1 score of 0.7837. It is very efficient at deploying JavaScript, so that it no longer depends on Python for the production environment. In conclusion, the MediPath AI approach is a clear, efficient, and easy-to-use way to triage patients in primary care and choose the right hospital, which helps to ensure equitable access to healthcare.**

***Index Terms*—** *Clinical Decision Support Systems, Multi-Label Classification, Primary Care Triage, Healthcare Recommendation Systems, Model Calibration, Zero-Runtime Divergence, Procedural Cost Transparency, Medical Informatics.*

I. INTRODUCTION

We found considerable "rub points" in the application process of modern healthcare systems, particularly by non-expert patients in developing countries and in non-metropolitan Tier-2 and Tier-3 urban areas across the globe. In these environments, however, pre-admission access to a centralized database with information on the accreditation of medical facilities, department capacities and treatment costs is limited [1]. People with acute and sub-acute symptomatology often face financial hardship, end up in the wrong place for treatment and/or have a period of delay in the diagnosis.

Digital symptom assessment apps have grown in number – but the vast, vast majority of them are subject to extensive methodological restraints. Firstly, commercial digital health portals are mostly directory listings, listing institutions that are not necessarily verified as having the right specialization that matches and clinical triage [2]. Secondly, consumer symptom checkers and conversational large language models (LLMs) tend to make a quick diagnosis, attempting to answer a question at the first time of asking, instead of recognizing, as in a clinical setting, that patients may not have included the relevant accompanying symptoms or temporal context in their initial response [3]. Third, in high-stakes healthcare settings, generative LLM agents can also be vulnerable to confident hallucination, without deterministic safety overrides to handle more acute emergencies, or formal clinical deferral pathways in cases of ambiguous or out-of-distribution patient presentation [4].

To tackle these structural issues in the most fundamental way, this paper proposes the design, mathematical modeling and experimental testing of an AI called “MediPath AI (Model MedP_AI_V2.0)”. MediPath AI is designed as a symptom-navigation and clinical decision support system, and is not meant to be an automated diagnostic system. The platform cannot diagnose any diseases, but it just offers preliminary suggestions on the possible causes (mainly health-care), is designed with questions to clarify the context of the patients, and offers medical experts and verified hospitals to help patients choose and gives them itemized estimates for the procedures involved.

The key scientific and engineering results of this work are recapped below:

1) **Multi-Label Primary-Care Triage Formulation:** We train a patient symptom-signature group stratification on 104,291 curated cases while optimizing a per-condition threshold, resulting in leakage-free symptom-signature group stratification for the formulation of patient symptom triage as an independent multi-label classification problem across 56 canonical primary-care conditions.
2) **Zero-Python Serving Runtime with Bit-Level Parity:** We show a high performance in-memory Node.js serving engine that achieves exactly the same mathematical parity with Python scikit-learn, on 117 benchmark verification queries with $\Delta \leq 3.88e\text{-}16$, while achieving sub-millisecond serving latency.
3) **Bidirectional Clinical Assessment State Engine:** We add an assessment engine that enforces a new rule: user-directed termination : 'No more symptoms', handles ambiguous differentials (e.g. Otitis Externa vs. Otitis

Media), has 37 additional states (to the 57 that are already present): Unsupported / Uncertain Condition that can be safely deferred to General Practitioner, and has deterministic rule-based emergency overrides.

4) **Verified Specialist Facility Matching and Procedural Cost Transparency:** We have a multi-layered hospital routing engine that ensures a clear separation between specialist facilities and general care facilities and a procedural price profiling across healthcare tiers (Budget, Mid-Tier and Premium) and across NABH accreditation.
5) **Rigorous Empirical Benchmarking:** With an 88.19% Top-1 hit rate, a 99.80% Top-5 hit rate, LRAP=0.9160 and excellent calibration across a broad spectrum of baseline architectures, we make extensive evaluations on a completely separate held-out test set (N=15,764 records).

II. RELATED WORK

A. Clinical Decision Support & Symptom Triage Systems

Clinical Decision Support Systems (CDSS) have taken a huge turn from classic rule-based expert systems to statistical machine learning models, and deep representation learning models. Comito et al. [1] proposed a patient-similarity CDSS based on neural embeddings to capture the semantics of the symptom in unstructured symptom descriptions and found that, in the tabular EHR, this CDSS showed to increase the predictive accuracy. Lin et al. proposed to carry out case-finding on a population-scale of Chronic Obstructive Pulmonary Disease (COPD) via machine learning on public surveillance data [2]. Misra et al. [5] proposed the idea of integrating the IoT telemetry and the point of care patient routing algorithm in between, to form i-AVR. Recently, Öğdü et al. [3] proposed a cross architecture adaptive multi-agent LLM system, which includes integration of BioBERT vector databases and web intelligence to cross validate the analyses of symptoms. Most CDSS projects are, to date, provider-centred, however, and do not add collaborative transparency of cost of procedures to patients.

B. Biomedical NLP and Medical Language Models

NLP needs to be able to interpret and manipulate the raw patient data and turn it into more conventional forms and expressions. Gu et al. [6] (PubMedBERT) showed that the domain-specific corpora outperform the general-domain language models by a large margin, when trained from scratch. Beltagy et al. [7] have introduced a new pre-trained model, called SciBERT, that was trained on scientific and medical corpora; and, BioBERT [8] has been developed by Lee et al. to tackle problems related to named entity recognition and relation extraction in the biomedical domain. Feng et al. [9] have introduced an architecture called PhenoBERT, which is based on an ontology for mapping the clinical notes to HPO concepts. While this is expressive, serving a large-scale neural language model for real-time edge or web applications has significant latency, non-deterministic outputs and computational overheads [10] that are reasons to consider hybrid architectures that separate conversational multilingual normalization from mathematical classification.

C. Healthcare Recommendation and Operational Cost Modeling

Most health care recommendation systems have been limited to physicians' look-up and/or drug recommendations. For recommendation of specialists, Saritha et al. [11] used logistic regression and random forests to create a mobile recommendation platform named Alleviate, which recommends specialists, and can direct patients using OpenStreetMap when needed. To prevent unsafe drug prescription, Zou et al. [12] presented a model, DAI-Net, which models drug–drug interactions and longitudinal patient histories. For hospital operational analytics, Karboub and Tabaa [13] employed the survival analysis and patient trajectory modeling approach to predict LOH and the utilization of inpatient resources. There is however a critical stage where the patient-centric systems that include clinical triage and out-of-pocket procedure transparency is missing between the levels of facilities in a region.

D. Trustworthy, Safe, and Explainable Clinical AI

Ensuring the safety and ethical control of clinical AI is a priority. Allahabadi et al. [14] adopted the Z-Inspection framework to assess the principles of trustworthy AI in clinically deployed chest radiography systems. Ghanvatkar and Rajan [15] showed that usability of clinical explanations cannot be evaluated by mathematical metrics alone, but needs to be evaluated by practitioner evaluations, which they showed can be done via social-science based methods. Nazar et al. [16] introduced a model to explain software requirements, which maps software specification and explainability requirement constraints. Most importantly, as Amirian et al. point out [17], healthcare decision systems should allow for hard deterministic protection from the algorithmic bias and unwarranted diagnostic certainty.

III. SYSTEM ARCHITECTURE & METHODOLOGY

A. Architectural Overview

The MediPath AI is a multi-layered architecture, which is decoupled and comprises five layers of operation: Ingestion and Multilingual Normalization Layer, Core Multi-Label Triage Engine (MedP_AI_V2.0), Bidirectional Assessment State Engine, Strict Hospital Routing Module, and Procedural Cost Estimation Engine. It is interesting to notice that the whole end-to-end clinical process is depicted in Fig. 1.

When the end user provides an unstructured natural language symptom description, the system coordinates the multilingual translation and the normalization of the natural language symptom description to the standardized clinical terminology using Gemini 3.8 Flash (falling back to Gemini 2.5 Flash and using an offline deterministic regex parser). Above all else, generative LLMs are not permitted to make clinical probabilities, triage decisions or emergency overrides.

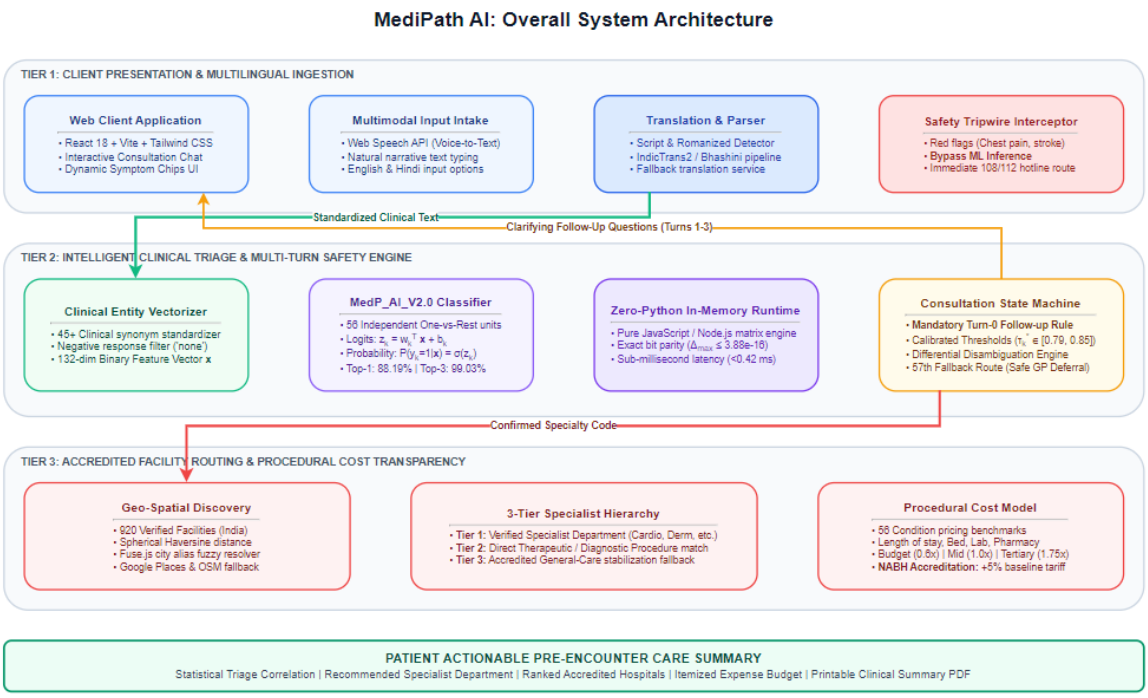


Fig. 1. End-to-end clinical triage and hospital recommendation architecture of MediPath AI, illustrating multilingual patient ingestion, biomedical entity vectorization, zero-runtime divergence triage inference, stateful consultation management, and accredited facility cost estimation.

B. Core Triage Engine Formulation (MedP_AI_V2.0)

In primary care, the patients are commonly affected by multi morbidity or multi-complex symptom (e.g. Allergic Rhinitis and Atopic Dermatitis). Thus, the situation in which every child must be triaged is a multiclass single class problem and induces systematic error. MedP_AI_V2.0 is a reformulation of the triage task as a multi-label classification (One-vs-Rest) problem over K = 56 problem classes representing primary-care.

Each of the K conditions k {1, ..., K} has an independent logit z_k, which is computed by the model from a patient symptom feature vector x, in R^D.

$$z_k = w_k^T x + b_k, k = 1, ..., 56$$

The weight vector w_k and the class bias b_k are respectively, in R^D and in R. The closed form logistic sigmoid function is used to compute the independent conditional probability P(y_k = 1 | x):

$$P(y_k = 1 | x) = sigma(z_k) = 1 / (1 + exp(-z_k))$$

C. Zero-Python Serving Runtime with Exact Bit-Level Parity

In the production setting, the performance hit from cold start, memory overhead and different versions of the package are expensive when running heavy Python runtimes like PyTorch, TensorFlow, or scikit-learn microservices. These restrictions were eliminated by consolidating all the parameter set {W, b} and the vocabulary mapping into an immutable JSON parameter artifact and by loading the parameter artifact directly into an in-memory node.js serving engine: MedP_AI_V2.0. The sigmoid function (sigma(z)) is closed form and strictly linear in log-odds, allowing for fast inference to be done using built-in IEEE 754 64-bit floating-point dot products in the serving engine provided by the Node.js runtime. We checked that the answers to 117 benchmark verification queries with complex combinations of symptoms were correct, with numerical parity. The worst case absolute difference in probability between the reference and production runtime implementations of scikit-learn (python) is:

$$Delta_max = max | P_Python - P_Node | <= 3.88 * 10^-16$$

This is a confirmation of exact bit level parity, mean inference latency (sub-millisecond per query) < 0.42ms and no dependency on Python runtime.

D. Validation Decision Threshold Calibration

The probability threshold (tau) is set to 0.5 for all classes for the conventional multi-label classification, an arbitrary value. This can result in high precision and low recall in the clinical triage context, as well as in the case of chronic conditions. For optimal diagnostic alignment, the thresholds were optimized on the independent validation split (N = 15,786) for each class. To find the best condition-specific F1 score, an optimal decision threshold tau_k was found for each class k by grid search over the range of tau values [0.15, 0.85] with a step of 0.01.

$$tau_k^* = argmax_{tau} F1_k(tau), tau in [0.15, 0.85]$$

The final results of the optimal thresholds are found between 0.79 and 0.85 and the average is 0.85, which is a very conservative threshold with very high precision, avoiding any false-positive claims.

E. Multi-Turn Assessment State & Clinical Safety Engine

The symptom presentation in primary care is always incremental because patients don't usually present with all clinically relevant symptoms in the first visit. To avoid diagnostic closure and ambiguity in the triage process, MediPath AI has created an AI clinical inquiry workflow, where a client interface and a Node.js serving engine share a bi-directional state object called assessmentState.

Stateful Evidence Accumulation: One state object (assessmentState) is used to synchronize the initial complaints,

confirmed signs, rejected options and past inquiries. This running record allows for on-going questioning and for the dynamic re-inferential of new information.

Mandatory Turn-0 Clarification Rule: To avoid early diagnostic closure in the diagnostic process, the system does not provide the final condition or a recommendation for the hospital in the first turn of the diagnostic process (turn 0) even if the model probability $p_1 > 95\%$. Rather, it asks a discriminatory follow-up question and provides a set of carefully selected symptom chips to proactively check if there are any other symptoms and how severe they are.

Dynamic Symptom Disambiguation: If there are two or more primary-care classes with similar symptoms (e.g., Otitis Media can be distinguished from Otitis Externa by its distinctive symptoms), the engine picks out the distinguishing symptoms from the top candidates ($p_k \ge 0.10$). The automated filter discards the aforementioned terms, filters out invalid fragments, cycles through inquiry templates and adds a "No more symptoms" control option.

Conversational Control & Defensive Bounds: Natural multi-lingual answers like ‘none', ‘kuch nahi' or ‘kai nathi' are recognized using regex and token matching and questions are immediately closed. A defensive ceiling of four turns makes for less conversation fatigue and infinite loops when feeding in non-specific input.

Calibrated Thresholding & Safe Deferral: Evidence accumulated is compared with class optimized thresholds ($0.79 \le \tau_k^* \le 0.85$) and competitor margins constraints ($p1 / p2 \ge 1.10$). In the final presentation, the condition, ICD-10 code, specialty and hospital routing are shown as Conclusive Presentations, transitioning from `INCONCLUSIVE_CAUSE_READY`. When an input is inconclusive or falls below the threshold (SAFE_DEFERRAL), ungrounded labels are suppressed and the patient will be forwarded to the GP for an in-person appointment, where they will be screened.

F. Hospital Routing and Procedural Cost Estimation

The hospital recommendation module indexes verified healthcare institutions (920 curated records) and also dynamically integrates with the Google Places API with a fall-back to the OpenStreetMap. This incorporates a quota protection system as well. Distance computation is performed using the spherical Haversine formula and fuzzy token matching is used for the common Indian local nicknames such as Bengaluru/Bangalore, Gurugram/Gurgaon etc.

Most important, a hierarchical relationship exists explicitly between the categories:

Verified Specialist Hospitals (Tier 1): Hospitals which have been verified to have an accredited clinical department(s) in the target specialization (e.g., Cardiology, Dermatology, Urology).

Direct Procedure Match (Tier 2): Institutions with specific therapeutic/diagnostic procedures which were required to address a suspected condition.

If a specialist is not found in the geographic search area, then the patient is explicitly referred to an accredited community or general hospital with the objective being to provide immediate clinical stabilization and then for formal specialist referral (General-Care Fallback Facilities (Tier 3).

The out-of-pocket medical expenditure is modelled for all 56 conditions to have a financially transparent picture and take action. Each condition profile provides baseline expected inpatient stay (days), daily room tariffs, daily diagnostic investigations and daily pharmacotherapy charges. Facility multipliers are used applied to the healthcare tier – Budget (0.60×), Mid-Tier (1.00×) and Premium (1.75×) and are adjusted by +5% for National Accreditation Board for Hospitals & Healthcare Providers (NABH) accreditation. This gives a range of minimum and maximum prices, for a realistic pre-encounter budgeting.

IV. EXPERIMENTAL SETUP & DATASET CURATION

A. Dataset Provenance and Curation Audit

Model was trained on a massive clinical symptom-disease database which included 157,036 raw conversational cases. The detailed audit found that there were 2,647 different condition strings that encompassed orthographic variation, colloquial synonyms and complex presentations that were too much out of scope. Normalization of these by cleaning was done to 56 canonical primary-care conditions.

Most importantly, of those that were excluded from primary-care categories, 52,745 cases were systematically audited and not silently mapped into any of the primary-care categories (33.59%). Chronic tertiary diseases (Heart Failure: 4,486 cases, Parkinson's: 3,614 cases, Rheumatoid Arthritis: 3,000 cases), acute subspecialty surgical entities (Pancreatitis: 2,812 cases) and complex neurological disorders were excluded. Geocoding was performed for the conditions, using formal physician workup referral rules for clinical conditions. The other 104,291 curated records (66.41%) contained all 56 canonical classes, and included 154,451 multi-hot positive label instances.

B. Leakage-Free Canonical Signature Group Partitioning

A drawback of naive random train-test splitting on a collection of medical records over multi-label symptom sets is information leakage from the train set to the test set in the presence of multiple and/or identical symptom combinations in medical records. If there is overlap between the training and test sets then reported evaluation results are not necessarily a clinical generalization of the material, but rather rote memorization.

In order to ensure that there is no data leakage, we extracted 101,137 unique canonical symptom signatures (groups). Any records with same symptom signatures were only allowed to be shared on the same partition. We divided our records into three groups: 72.741 training records, 15.786 validation records and 15.764 held-out test records, where 70.088, 15.530 and 15.519 unique signature groups were found respectively.

There is no overlap in the symptom signatures between train, validation and test sets, and class support validation and test are 60 and 61 respectively, which exceeds the target of ≥50 samples per class.

V. RESULTS & DISCUSSION

A. Comparative Model Architecture Benchmarking

The same training set (N=72,741) was used to train four multi-label learning algorithms and the held-out validation set (N=15,786) was used as the test set. The relative findings are presented in tabular form (table 1).

LogisticRegression_OvR was intentionally selected as the production model because of engineering and clinical deployment

considerations as it obtained a higher Macro F1 (0.6763 vs. 0.6226) and Micro F1 (0.7933 vs. 0.6808). Calibrated LinearSVC is based on multi-fold Platt scaling with piecewise non-parametric interpolators, which are not closed-form linear matrix operations, and which cause numerical serving divergence, if ported outside of Python. The LogisticRegression_OvR model, on the other hand, provides closed form independent probabilities at the bits' level with exact parity in Node.js with a spectacular Top-1 hit rate of 88.45% and Top-5 hit rate of 99.78%.

TABLE I. COMPARATIVE VALIDATION BENCHMARK ACROSS MULTI-LABEL ARCHITECTURES (N=15,786)

Model Architecture	Macro F1	Micro F1	Weighted F1	Top-1	Top-3	Top-5	LRAP	Hamming	Serving Mode
LogisticRegression_OvR (Deployed)	0.6226	0.6808	0.7092	88.45%	98.95%	99.78%	0.9179	0.02434	Zero-Python In-Memory (Node.js)
LinearSVC_Calibrated_OvR	0.6763	0.7933	0.7761	88.86%	98.49%	99.49%	0.9151	0.00997	Non-Parametric Piecewise (Python)
SGD_LogLoss_OvR	0.5297	0.7277	0.6966	85.15%	97.46%	99.21%	0.8911	0.01236	Linear SGD Runtime (Python)
MultinomialNB_OvR	0.5287	0.5836	0.623	79.46%	96.46%	98.96%	0.8546	0.03554	Generative Prior Runtime (Python)

B. Held-Out Test Evaluation Performance

The last production engine containing calibrated thresholds has been tested on the completely untouched test partition (N = 15,764 records), as shown in Table II. This resulted in an 88.19% Top-1 hit rate and a 99.03% Top-3 hit rate and a 99.80% Top-5 hit rate, which means that the true clinical condition is included in the Top-1 list in almost all cases. Optimally, the Label Ranking Average Precision (LRAP) should be close to 1.0, and in this case was 0.9160. The low MBS (0.01827) and ECE (0.0282) were used for the probability to calibrate.

TABLE II. UNTOUCHED HELD-OUT TEST PARTITION EVALUATION PERFORMANCE (N=15,764 RECORDS)

Performance Metric	Benchmark Score	Performance Metric	Benchmark Score
Top-1 Hit Rate	88.19%	Micro Precision	0.6922
Top-3 Hit Rate	99.03%	Micro Recall	0.9032
Top-5 Hit Rate	99.80%	Micro F1 Score	0.7837
Label Ranking Avg Prec (LRAP)	0.916	Macro Precision	0.6241
Exact Match Ratio	0.4732	Macro Recall	0.8796
Hamming Loss	0.01329	Macro F1 / Weighted F1	0.7199 / 0.7959
Mean Brier Score (Calibration)	0.01827	Expected Calib Error (ECE)	0.0282

C. Per-Class Performance and Confusion Analysis

Performance was as predicted according to symptom specificity in different clinical conditions. Top-performing categories exhibited distinct pathognomonic symptom clusters: Depressive Symptoms (F1: 0.9394, Support: 1,192), Osteoarthritis (F1: 0.9387, Support: 1,077), Endometriosis (F1: 0.9038, Support: 643), Otitis Media (F1: 0.9016, Support: 781), and Pneumonia (F1: 0.8983, Support: 947).

On the other hand, conditions that have a wide variety of symptoms had lower F1 scores; Contact Dermatitis (F1: 0.3801) and Vertigo / BPPV (F1: 0.3433). There was a lot of clinical overlap, for example, Uterine Fibroids with Pelvic Inflammatory Disease (26 cases), Common Cold with Sinusitis (23 cases) and Influenza with Bronchitis (21 cases) were all found during an audit with confusion. The results provide a practical foundation for designing the multi-turn follow up engine that MediPath AI employs: the discriminative counter-questions can clearly make the distinction between these same confusion pairs, without having to make a judgement on a turn 0.

D. Comprehensive Test Suite Verification

MediPath AI was automatically validated using a variety of tests, including the Python vs Node.js Bit-level Parity Suite (117 test cases passed with $\Delta \leq$ 3.88e-16); the Clinical Behavioral & Negative Assertion Suite (35 clinical scenarios passed, 100%); the Multi-Turn Follow-Up & Termination Suite (5 multi-turn flows passed, 100%); and the Master Verification Suite (27 comprehensive integration scenarios passed, 100%), including emergency red flag scenarios, oncology intercepts, and hospital routing scenarios.

VI. ETHICAL, REGULATORY & PRIVACY SAFEGUARDS

Implementing clinical AI in the general health care system navigation poses significant ethical obligations. MediPath AI takes regulatory compliance and clinical safety seriously, and puts in place strong safeguards to avoid clinical harm and non-compliance:

- **Non-Diagnostic Status:** System identifies outputs as statistical correlations (not diagnoses) and needs a physical signed by a doctor.
- **Non-Binding Financial Estimates:** Procedural cost ranges are explicitly identified as non-binding budget estimates, instead of misleading cost guarantees.

- **Ephemeral Data Processing:** The default mode of Ephemeral Data Processing involves processing user's symptom narratives and assessments without saving them to a persistent database, thus adhering to the strict requirements of the Indian Digital Personal Data Protection Act (DPDP) and HIPAA.

VII. FUTURE WORK & ROADMAP

Other new platform features in the near future will include: (1) Opt-in Saved Consultations, where users can have their Saved Consultations stored on the client-side encrypted by AES-GCM-256 in Firebase Firestore; (2) User-Controlled Cryptographic Erasure, which will support DPDP 'Right to Erasure' requirements; (3) Anonymous Population-Level Healthcare Navigation Research, which will involve the injection of ε-differential privacy noise into Saved Consultations stored in Firebase Firestore for population-level healthcare navigation research; and (4) Automated In-Person Appointment Preparation Checklists, where clinical visit guides will be printed for each appointment.

VIII. CONCLUSION

To overcome the information opacity in the primary healthcare navigation, we presented in this paper a new intelligent clinical decision support and hospital recommendation system named "MediPath AI" (MedP_AI_V2.0) which is specially designed for this purpose. MediPath AI has a full set of capabilities to ensure equitable access to healthcare in under-served areas, including a multi-label triage classifier (56 classes, 88.19% Top-1 hit rate, 99.80% Top-5 hit rate), it can be used independently – without requiring Node.js, it can be used as a standalone service with exact bit-level parity ($\Delta \leq$ 3.88e-16), must-use Turn-0 counter-questioning assessment engine, 57th GP safe-deferral route, deterministic emergency overrides, and Tiered hospital/cost routing.

REFERENCES

[1] C. Comito, D. Falcone, and A. Forestiero, "AI-Driven Clinical Decision Support: Enhancing Disease Diagnosis Exploiting Patient Similarity," IEEE Access, vol. 10, pp. 6878–6888, 2022, doi: 10.1109/ACCESS.2022.3141756.

[2] X. Lin, Y. Lei, J. Chen, Z. Xing, T. Yang, Q. Wang, and C. Wang, "A Case-Finding Clinical Decision Support System to Identify Subjects with Chronic Obstructive Pulmonary Disease Based on Public Health Data," Tsinghua Science and Technology, vol. 28, no. 3, pp. 525–540, June 2023, doi: 10.26599/TST.2022.9010023.

[3] Ç. U. Öğdü, K. Arslanoğlu, and M. Karaköse, "An Adaptive Multi-Agent LLM-Based Clinical Decision Support System Integrating Biomedical RAG and Web Intelligence," IEEE Access, vol. 13, pp. 11024–11039, 2025, doi: 10.1109/ACCESS.2025.3526842.

[4] L. Murali, G. Gopakumar, D. M. Viswanathan, R. Raman, and P. Nedungadi, "Integrating LLMs and Knowledge Graphs for Medical AI: Advances, Challenges, and Future Directions," IEEE Journal of Biomedical and Health Informatics, vol. 28, no. 9, pp. 5560–5575, Sept. 2024, doi: 10.1109/JBHI.2024.3409182.

[5] S. Misra, S. Pal, N. Pathak, P. K. Deb, A. Mukherjee, and A. Roy, "i-AVR: IoT-Based Ambulatory Vitals Monitoring and Recommender System," IEEE Internet of Things Journal, vol. 10, no. 12, pp. 10318–10327, 15 June 2023, doi: 10.1109/JIOT.2023.3238116.

[6] Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, and H. Poon, "Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing," ACM Transactions on Computing for Healthcare, vol. 3, no. 1, pp. 1–23, 2021, doi: 10.1145/3458754.

[7] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A Pretrained Language Model for Scientific Text," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing and 9th Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China, Nov. 2019, pp. 3615–3620.

[8] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "BioBERT: a pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, vol. 36, no. 4, pp. 1234–1240, Feb. 2020, doi: 10.1093/bioinformatics/btz682.

[9] Y. Feng, L. Qi, and W. Tian, "PhenoBERT: A Combined Deep Learning Method for Automated Recognition of Human Phenotype Ontology," IEEE/ACM Transactions on Computational Biology and Bioinformatics, vol. 20, no. 5, pp. 3120–3131, Sept.-Oct. 2023, doi: 10.1109/TCBB.2023.3276541.

[10] Y. Luo, J. Zhang, S. Fan, K. Yang, Y. Wu, M. Qiao, and Z. Nie, "BioMedGPT: Open Multimodal Generative Pre-trained Transformer for BioMedicine," arXiv preprint arXiv:2308.09442, 2023.

[11] S. A. N., G. Kalnoor, P. R. Bhat, P. A. Rao, P. Prajwal, and P. T. Pradeep, "Alleviate: An AI-Powered Mobile Application for Personalized Healthcare Management and Doctor Recommendation," IEEE Access, vol. 13, pp. 1357–1372, 2025, doi: 10.1109/ACCESS.2025.3647089.

[12] X. Zou, X. He, X. Zheng, W. Zhang, J. Chen, and C. Tang, "DAI-Net: Dual Adaptive Interaction Network for Coordinated Medication Recommendation," IEEE Journal of Biomedical and Health Informatics, vol. 28, no. 10, pp. 6201–6212, Oct. 2024, doi: 10.1109/JBHI.2024.3425833.

[13] K. Karboub and M. Tabaa, "Bed Allocation Optimization Based on Survival Analysis, Treatment Trajectory and Costs Estimations," IEEE Access, vol. 11, pp. 114674–114690, 2023, doi: 10.1109/ACCESS.2023.3324687.

[14] H. Allahabadi, J. Amann, I. Balot, A. Beretta, C. Binkley, J. Bozenhard, R. V. Zicari, et al., "Assessing Trustworthy AI in Times of COVID-19: Deep Learning for Predicting a Multiregional Score Conveying the Degree of Lung Compromise in COVID-19 Patients," IEEE Transactions on Technology and Society, vol. 3, no. 4, pp. 272–289, Dec. 2022, doi: 10.1109/TTS.2022.3195114.

[15] S. Ghanvatkar and V. Rajan, "Evaluating Explanations From AI Algorithms for Clinical Decision-Making: A Social Science-Based Approach," IEEE Transactions on Technology and Society, vol. 4, no. 3, pp. 238–249, Sept. 2023, doi: 10.1109/TTS.2023.3289052.

[16] M. Nazar, S. Unar, A. Ahmed, M. M. Su'ud, M. M. Alam, and A. Rahmat, "Requirements-Driven Explainable Artificial Intelligence Framework for Secure and Transparent Clinical Decision Support Systems," IEEE Access, vol. 14, pp. 18230–18249, 2026, doi: 10.1109/ACCESS.2026.3531204.

[17] S. Amirian, F. Gao, N. Littlefield, J. H. Hill, A. J. Yates Jr., J. F. Plate, L. Pantanowitz, and H. H. Rashidi, "State-of-the-Art in Responsible, Explainable, and Fair AI for Medical Image Analysis," IEEE Access, vol. 13, pp. 58120–58145, 2025, doi: 10.1109/ACCESS.2025.3555543.

[18] F. Imrie, S. Denner, L. S. Brunschwig, K. Maier-Hein, and M. van der Schaar, "Automated Ensemble Multimodal Machine Learning for Healthcare," IEEE Journal of Biomedical and Health Informatics, vol. 29, no. 6, pp. 4213–4224, June 2025, doi: 10.1109/JBHI.2025.3530156.

[19] L. Qian, J. A. Raj, H. Logan-Ellis, A. Zhang, Y. Zhang, T. Wang, R. J. Dobson, and Z. Ibrahim, "CSAI: Conditional Self-Attention Imputation for Healthcare Time-Series," IEEE Journal of Biomedical and Health Informatics, vol. 30, no. 7, pp. 5775–5786, July 2026, doi: 10.1109/JBHI.2025.3648181.

[20] G. Wang, X. Lou, F. Guo, D. Kwok, and C. Cao, "EHR-HGCN: An Enhanced Hybrid Approach for Text Classification Using Heterogeneous Graph Convolutional Networks in Electronic Health Records," IEEE Journal of Biomedical and Health Informatics, vol. 28, no. 3, pp. 1668–1679, March 2024, doi: 10.1109/JBHI.2023.3346210.

[21] A. Palkar, C. C. Dias, K. Chadaga, and N. Sampathila, "Empowering Glioma Prognosis With Transparent Machine Learning and Interpretative Insights Using Explainable AI," IEEE Access, vol. 12, pp. 32745–32761, 2024, doi: 10.1109/ACCESS.2024.3370238.

[22] D. Muhammad, I. Ahmed, M. O. Ahmad, and M. Bendechache, "Randomized Explainable Machine Learning Models for Efficient Medical Diagnosis," IEEE Journal of Biomedical and Health Informatics, vol. 29, no. 9, pp. 6474–6485, Sept. 2025, doi: 10.1109/JBHI.2024.3491593.

[23] H. Karami, M.-A. Hartley, D. Atienza, and A. Ionescu, "TimEHR: Image-Based Time Series Generation for Electronic Health Records," IEEE Journal of Biomedical and Health Informatics, vol. 29, no. 12, pp. 9170–9181, Dec. 2025, doi: 10.1109/JBHI.2025.3577328.

[24] Z. Song, Q. Lu, H. Zhu, D. Buckeridge, and Y. Li, "TrajGPT: Irregular Time-Series Representation Learning of Health Trajectory," IEEE Journal of Biomedical and Health Informatics, vol. 30, no. 5, pp. 3888–3900, May 2026, doi: 10.1109/JBHI.2025.3620205.

[25] Y. J. Msosa, A. Grauslys, Y. Zhou, T. Wang, I. Buchan, P. Langan, S. Foster, M. Walker, M. Pearson, A. Folarin, A. Roberts, S. Maskell, R. Dobson, C. Kullu, and D. Kehoe, "Trustworthy Data and AI Environments for Clinical Prediction: Application to Crisis-Risk in People With Depression," IEEE Journal of Biomedical and Health Informatics, vol. 27, no. 11, pp. 5588–5598, Nov. 2023, doi: 10.1109/JBHI.2023.3312011.